Pradyumna Kaushal

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Education

National Institute of Technology (NIT), Tiruchirappalli **Trichy, India**
Bachelor of Technology (B.Tech.) (Final Year) *Aug. 2023 – May 2027*
Major in Production Engg. with Minor in Computer Science & Engg. ***Dept. Rank 1*** *— CGPA:* ***9.89/10.0***

Research Interests

Trustworthy distributed and cyber-physical systems; security, privacy, and verifiable computing for industrial and mobility systems; digital twins, digital threads, and lifecycle-aware data infrastructures; intelligent manufacturing, Industrial IoT, and immersive cyber-physical interfaces (AR/VR); simulation, optimization, and decision-making for complex engineered systems

Experience

Universität Leipzig Leipzig, Germany
Researcher – Information Systems – Prof. Rainer Alt *May 2026 – Present*

- Developed a **DLT research decision-support system** using **Open Knowledge Format (OKF)**, graph-structured knowledge representation, deterministic multi-stage retrieval, provenance-aware reasoning, and LLM synthesis to transform complex Design Science Research into traceable, reusable DLT design knowledge.
- Operationalized this design knowledge for **cross-organizational e-commerce**, architecting **smart-contract registries and distributed trust layers** that bind product, listing, seller, purchase, and review state through **GS1/ECLASS identities, verifiable claims, cryptographic commitments, and auditable provenance** across marketplaces.

King Abdullah University of Science and Technology (KAUST) Thuwal, Saudi Arabia
Visiting Student Researcher (VSRP) – CyberSaR – Prof. Ali Shoker *Dec. 2025 – Present*

- Researching **secure, private, and resilient cyber-physical systems** for connected mobility and V2X, combining **zero-knowledge proofs, cryptographic commitments, Merkle structures, signed attestations, revocation, selective disclosure, and lifecycle-aware state verification**.
- Built and benchmarked an end-to-end **verifiable-state prototype** with adversarial, scalability, and lifecycle testing; achieved **100% rejection of mutated verifier packages** and maintained **constant verifier-facing proof multiplicity** as committed history scaled to **65K records**.

National Institute of Technology (NIT), Tiruchirappalli Trichy, India
Research Assistant - Circular Economy & Sustainability - Prof. S. Prasanna Venkatesan *Aug. 2025 – Dec. 2025*

- Advanced a **Government of India funded** circular-economy project on **carbon accounting, life cycle assessment, and traceability** of refurbished laptops, aligned with **ISO 14040/14044** and the **GHG Protocol Product Standard**; contributed to a **Digital Product Passport** workflow using **12N QR-based device identities** for MSMEs.
- Developed **RefurbCarbon**, a component-level emissions framework quantifying **per-device CO_2e savings** from refurbishment using **OEM PCFs/LCI datasets** and **IDEMAT 2026**, modelling replacement components, logistics, workshop energy, and **Scope 1–3 emissions**; built a **Python/FastAPI** engine for integration with the project's **VBA/Excel-based diagnostic and certification tool**.

Matter Motor Works Bangalore, India
Intern – Connected Vehicle Data Systems *May 2025 – Aug. 2025*

- Engineered a **real-time connected-vehicle telemetry architecture** supporting sensor ingestion up to **250 Hz**, using **TimescaleDB, PostgreSQL, TypeScript/Express, Prisma, HMAC-authenticated APIs**, and time-series aggregation for battery health, mileage, charging cycles, warranty, and vehicle-lifecycle analytics.
- Designed **privacy- and sovereignty-aware vehicle data exchange** across OEM, service, insurance, and regulatory workflows using **GAIA-X, Catena-X, GDPR, JSON-LD**, and field-scoped access controls; implemented **one-time 10-minute JWT disclosures**, signed audit trails, and data-minimization mechanisms for cross-organizational mobility systems.

Spider R&D, NIT Tiruchirappalli Trichy, India
Head of Blockchain Research *Sep. 2024 – Present*

- Lead research on **zero-knowledge proofs, decentralized identity, protocol security, and verifiable distributed systems**, spanning **zk-SNARKs/STARKs, Circom, Noir, authenticated data structures, Merkle proofs**, privacy-preserving protocols, and security-oriented system design.
- Lead **technical partnerships, funding pitches, and accelerator engagements** with **Cedi (INR 10L+ proposal), Xion, Walrus Foundation, and Yellow Network**; advanced projects through technical reviews, RFP rounds, accelerator stages, and external R&D collaborations.

Projects

Temporal & Actionability Audit of ML Delivery-Risk Models Aug. – Sep. 2026

IC-AMMA 2026

- Developed a **temporal, leakage-aware CatBoost pipeline** on **43,396 USAID GHSC-PSM order lines** to predict **OTD failure**, using order-entry-only features, forward temporal splits, train-only normalization, calibration, and **Order-Number-clustered bootstrap inference**.
- Designed a **support-aware actionability audit** using relative-order-quantity features, real-context **ICE**, temporal transportability, and held-out quantity-band validation; improved future performance by **+0.019 ROC-AUC** and **+0.031 PR-AUC**, identifying only **5/35 products** with sufficient future support.

CosmoTrace – Permissioned Supply-Chain Ledger with MES/DT Anchoring Jul. – Aug. 2025

Cosmos SDK & Ignite CLI

- Developed a **permissioned blockchain** with **CometBFT**, implementing **ACL/PoA validators**, actor **DIDs**, and custom modules for auditable **orders, WIP, QC, and shipment state** with Merkle-proof verification.
- Integrated **MES/digital-twin telemetry** through **OPC-UA/Modbus**, batched state using **Ev-Stack rollups**, and exposed an **IBC gateway** for inter-organizational traceability and external anchoring.

Programmable Escrow & Fraud-Resilient Marketplace Payments May – Jul. 2025

HackOn with Amazon – Season 5

- Built a **Solana escrow protocol** in **Rust + Anchor** using **PDA vaults** and programmable settlement logic to automate post-return fund release and mitigate refund and settlement fraud.
- Engineered an **AWS event-driven backend** with **EventBridge, Lambda, Step Functions, and DynamoDB**; integrated **Amazon Fraud Detector** for **50K+ transactions/day** risk scoring at **95% accuracy**.

IoT-Based Predictive Maintenance of Rotating Machinery Jan. – Mar. 2025

Industrial IoT & Industry 4.0 Project

- Engineered an **ESP32-based industrial condition-monitoring prototype** for motors and pumps, integrating **temperature (DS18B20), vibration (SW-420), current, speed, and voltage sensing** into an edge-to-cloud telemetry pipeline; enabled remote equipment-state monitoring, threshold-based anomaly detection, and real-time operator alerts through an IoT application.
- Designed the predictive-maintenance analytics stack around **Butterworth/Kalman filtering, RMS, kurtosis, spectral features, and normalized time-series signals**, with classification, anomaly detection, and **Remaining Useful Life (RUL)** modelling as downstream extensions; validated abnormal temperature and vibration detection on a functional prototype costing approximately **INR 1.5–2K**.

Distributed Encountered-Type Haptics in VR Sep. – Nov. 2024

Smart India Hackathon 2024

- Built the **VR interaction and spatial-mapping layer** in **Unreal Engine**, using tracked-hand **vertical ray casting** to resolve virtual contact points, surface geometry, and target poses for physical haptic rendering.
- Integrated a **distributed multi-bot haptic system**, translating raycast contacts into real-time motion targets over **UDP** with **ArUco localization** and force-feedback inputs for synchronized visual–tactile interaction.

Awards & Achievements

2026 **Saxon Student Mobility Program (SSMP) Scholar** – fully funded 3-month research stay in Germany

2026 **1st Place – ETHGlobal HackMoney, Yellow SDK** (**$5K**): **Agentropolis**, an AI-based DeFi game

2026 **Academic Rank Holder – NIT Trichy Institute Day 2026** for academic distinction

2026 **Ram&Thaila NITT UG Research Fellowship** – Digital Product Passports for Circular Manufacturing

2025 Awarded the **$10K Solana Almighty Grant** for **medici.ac**, a blockchain scholarship platform

2025 **Prize Winner – Solana CTF**, Superteam India, for on-chain cybersecurity challenges

2025 **1st Rank Medal – NIT Trichy**, highest performer across the B.Tech. first-year cohort (**10.0/10.0**)

2025 **S. J. Chainlu Medal of Excellence** – 1st Rank Holder in Production Engineering, NIT Trichy